PC500 | Power Centre

High-pressure feed pumping with brine energy recovery on a shared drive.

The axial-piston motor recovers pressure energy from the RO reject stream and assists the pump drive. The integrated arrangement reduces separate energy-recovery components and oil-related maintenance at the pump, while retaining project-specific feed and membrane requirements.

Manufacturer parameter	Published value
Shaft Material	Duplex
Piston Material	Duplex
Housing Material	Duplex
Max. Operating Pressure	85 Bar
Min. Operating Pressure	30 Bar
Min. Inlet Pressure at Inlet Pressure Switch Port	3.5 Bar
Max. Operating Temp.	50°C
Mounting Direction	Horizontal
Operating Medium	Sea/Salt Water
Drinkable Water Production	Up to 500m³/day
Power Rating	See graphs

Design and environmental value

Duplex shaft, piston and housing materials are specified for saline duty. Product-water lubrication removes a pump lubricating-oil circuit. Integrated energy recovery can reduce motor input at the matched RO duty.

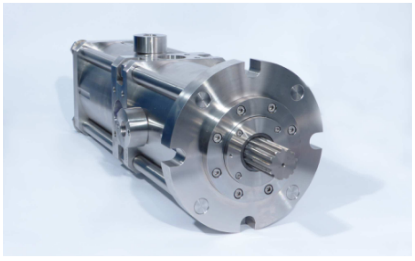
Measure the useful outcome

Compare total electrical kWh per m³ of acceptable product water at the same salinity, temperature, pressure and recovery. Include feed/boost pumps, pretreatment and auxiliaries in the stated boundary. No universal saving percentage is assumed.

Selection and procurement

State required permeate production and quality, feed salinity/temperature, membrane pressure, recovery ratio, operating hours, electrical supply and brine disposal conditions. Request the selected recovery code, approved motor/speed, materials and seals, complete arrangement and commissioning/service plan.

Source issues: the table lists 85 bar maximum; the circuit description lists 80 bar. Source installation speed: up to 1,250 rpm. Obtain the approved operating limit. Specific-energy charts label their axis kW/m³; obtain a confirmed unit before treating it as kWh/m³ performance.



Manufacturer unit photograph; motor/frame selected separately.

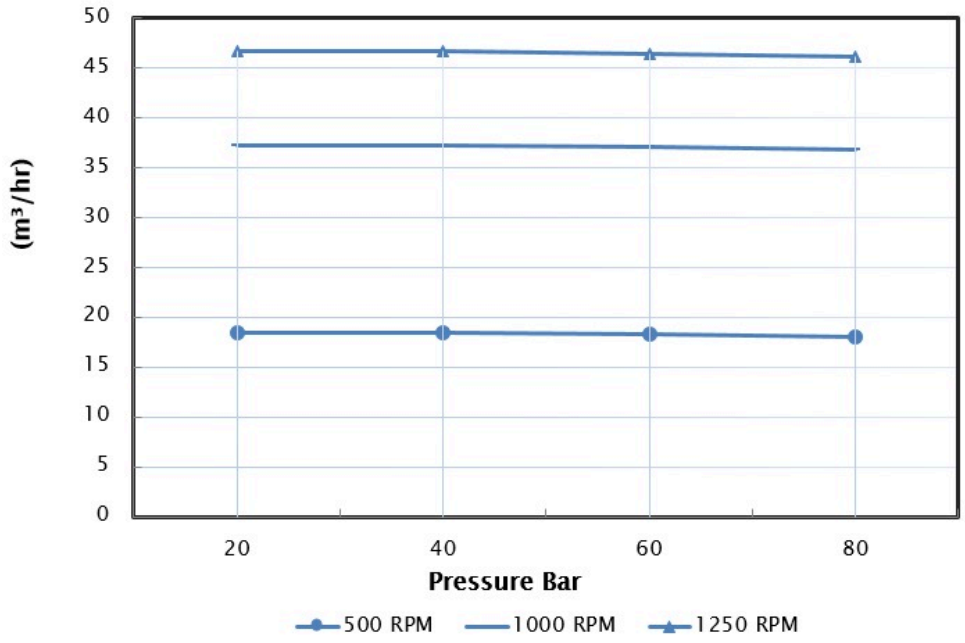
Recovery	Order code
30%	617-001
35%	618-001
40%	619-001
45%	620-001
50%	612-001

PC500 Power Centre V.01.25.pdf, pp. 2-4. Website product context checked 17 Oct 2026.

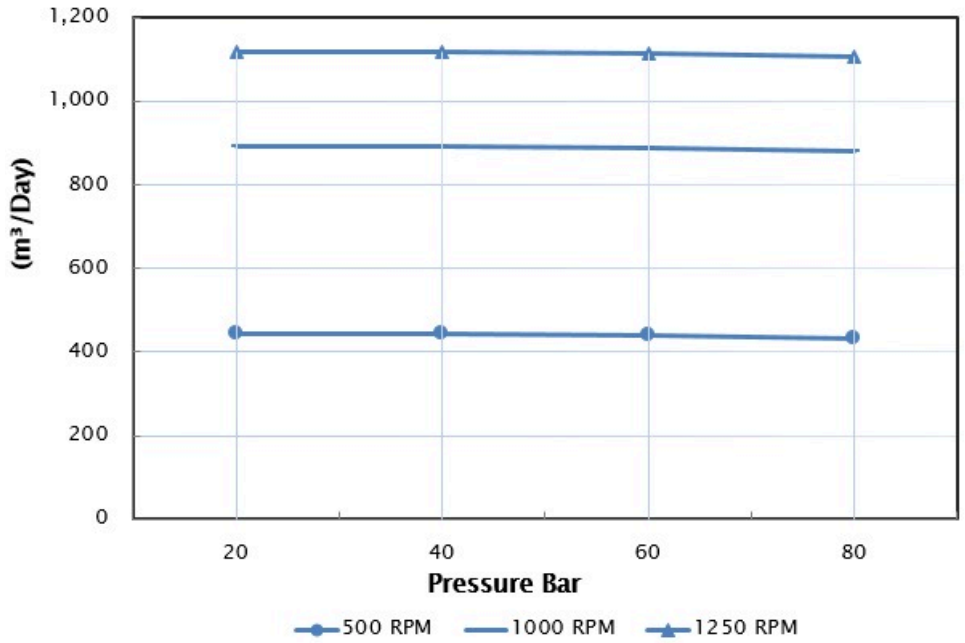
PC500 | Performance

Read the source legend and stated operating conditions for each chart.

Pump output flow chart, m³/h



Pump output flow chart, m³/day



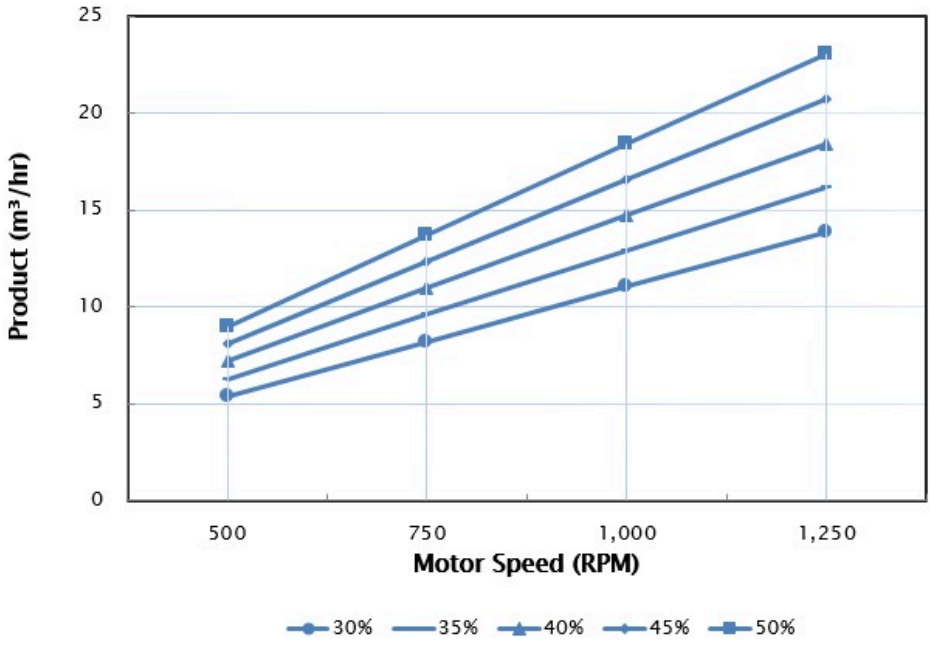
Production charts state 80 bar. Read speed and recovery from each legend. The source labels specific energy as kW/m³; obtain a confirmed unit and whole-plant energy basis before comparing offers.

PC500 Power Centre V.01.25.pdf, performance section. Manufacturer curves retained.

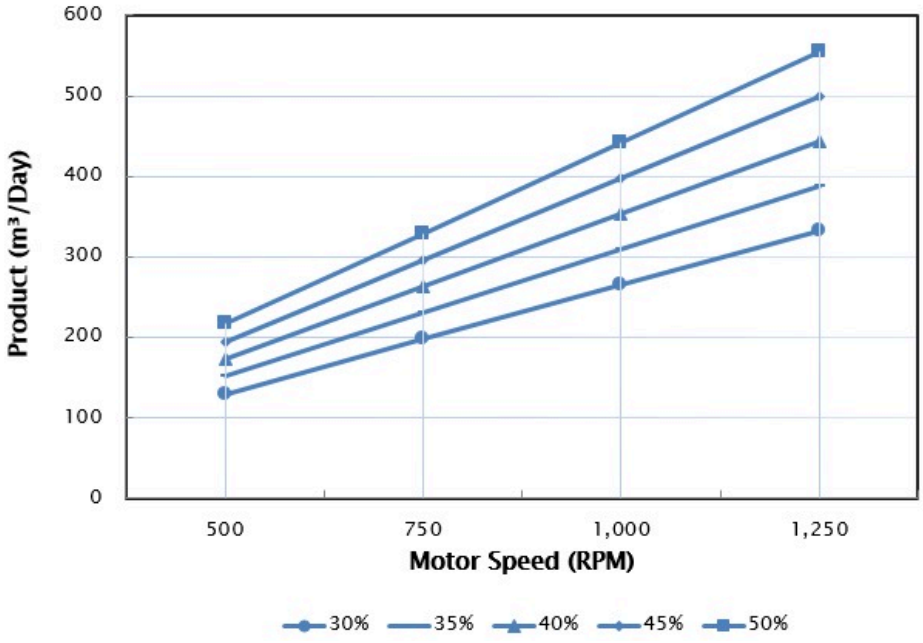
PC500 | Performance

Read the source legend and stated operating conditions for each chart.

Freshwater production chart, m³/h



Freshwater production chart, m³/day



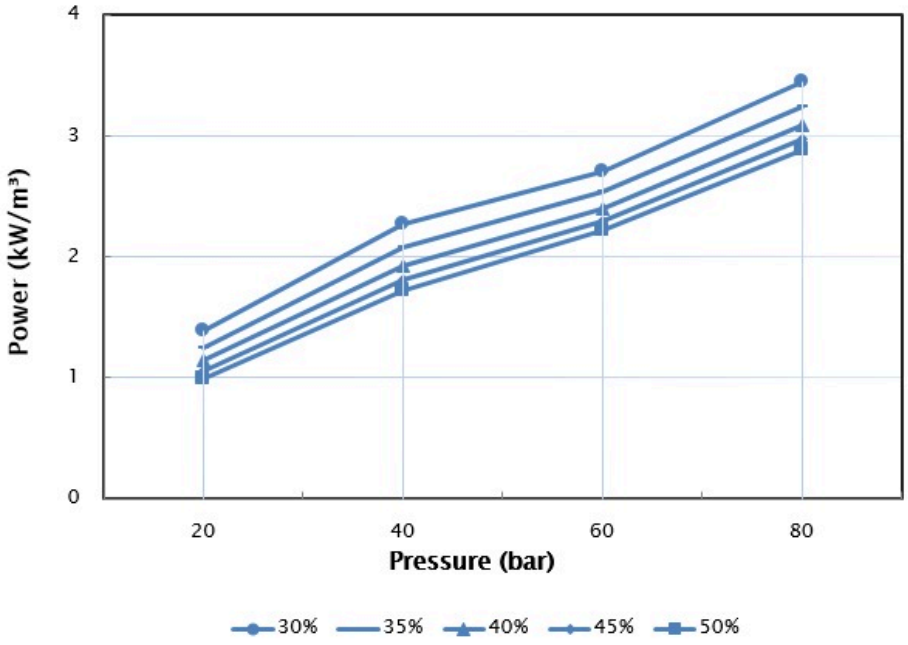
Production charts state 80 bar. Read speed and recovery from each legend. The source labels specific energy as kW/m³; obtain a confirmed unit and whole-plant energy basis before comparing offers.

PC500 Power Centre V.01.25.pdf, performance section. Manufacturer curves retained.

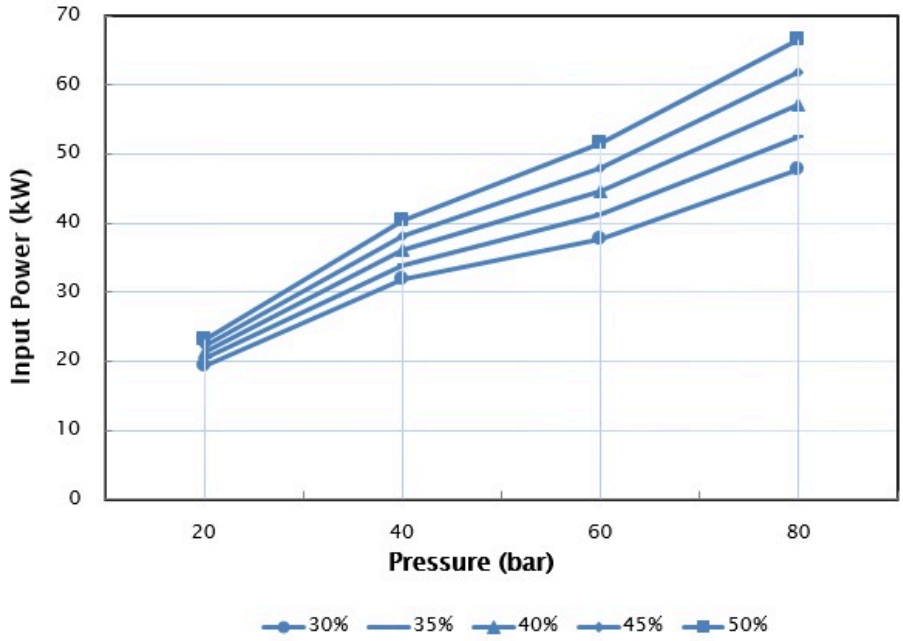
PC500 | Performance

Read the source legend and stated operating conditions for each chart.

Specific energy chart, Chart units: kW/m³



Input power chart, kW



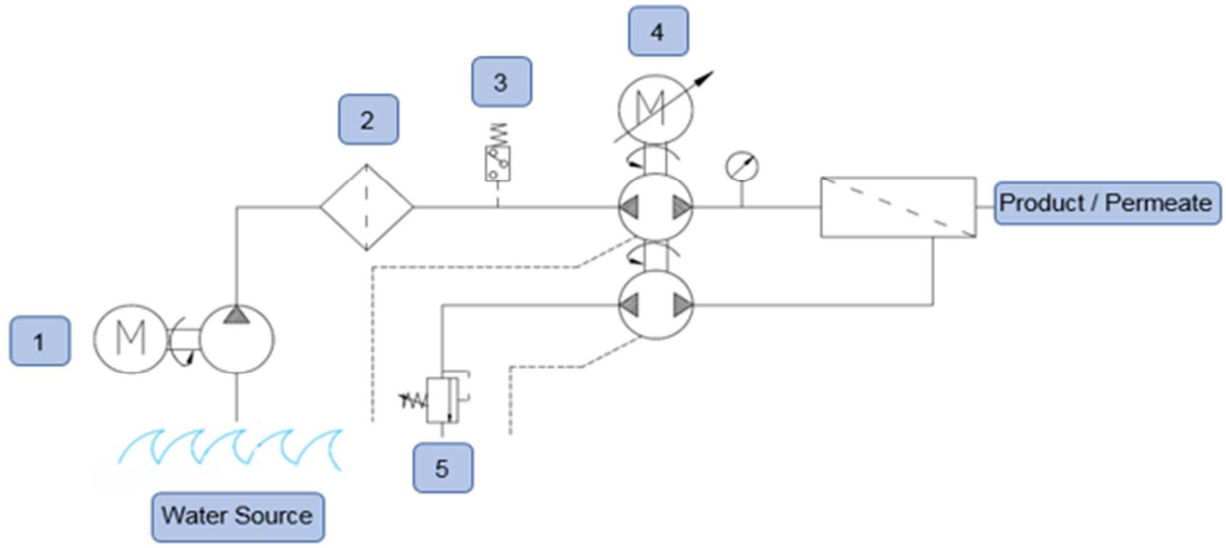
Production charts state 80 bar. Read speed and recovery from each legend. The source labels specific energy as kW/m³; obtain a confirmed unit and whole-plant energy basis before comparing offers.

PC500 Power Centre V.01.25.pdf, performance section. Manufacturer curves retained.

PC500 | Circuit & protection

Manufacturer technical reference | dimensions govern; reproduction is not to scale.

Circuit Diagram Example



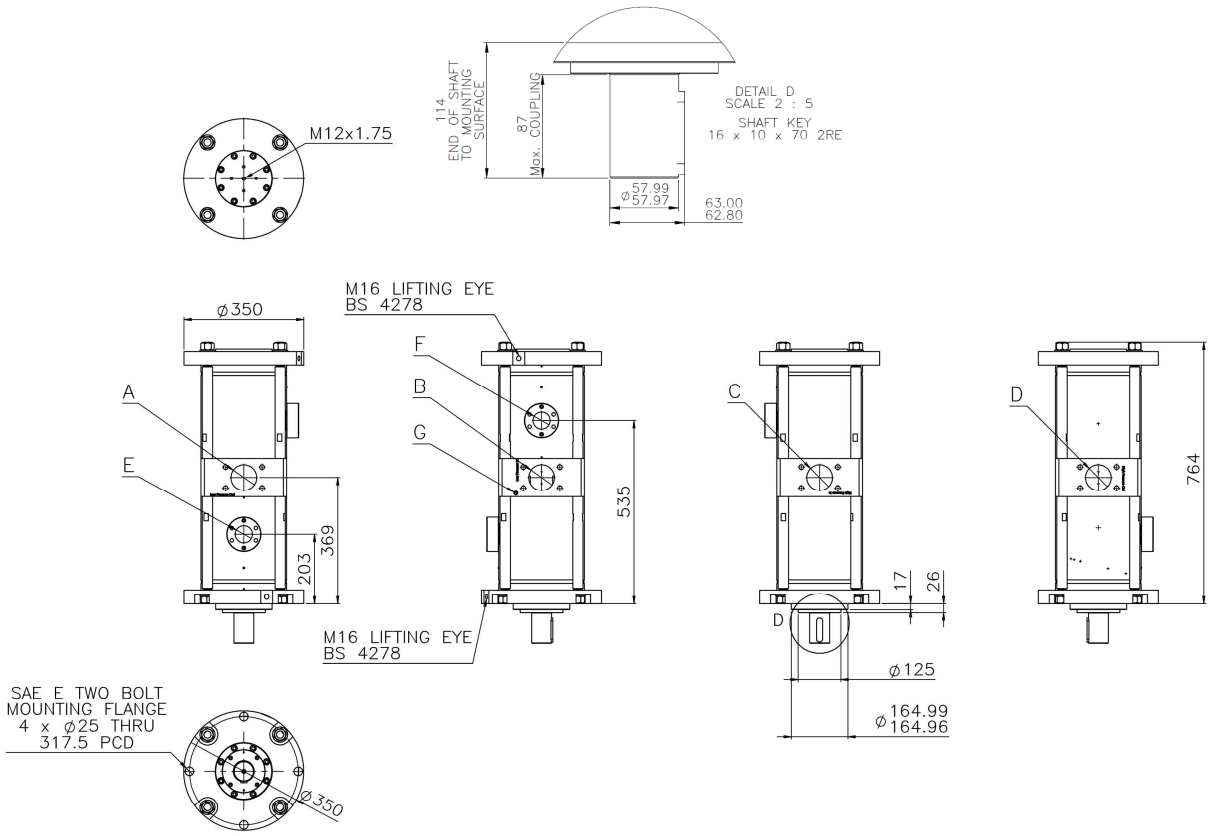
Item	Component	Function
1	Feed / Boost Pump	Ensures the inlet pressure supply to the pump's Low Pressure In connection at a minimum of 3.5 bar.
2	Filter	Ensure pumps minimum filtration requirements are met.
3	Inlet Protection Pressure Switch Set 3.5 bar	Ensures a continuous minimum 3.5 bar pressure at the pumps inlet. If the minimum of 3.5 bar is not achieved system shutdown to avoid pump damage.
4	Power Centre	The Water Hydraulics Power Centre including axial piston pump with integrated axial piston motor (ERD) and electric motor 500-1250 RPM, 80 bar maximum working pressure.
5	Back Pressure Valve	Setting the minimum backpressure on the Power Centre's Low Pressure Out

Separate case drains to a low-pressure sump/drain. The source circuit calls for a feed/boost pump and a minimum 3.5 bar (51 psi) pressure switch. Confirm the full P&ID and protection settings.

PC500 | Unit dimensions

Manufacturer technical reference | dimensions govern; reproduction is not to scale.

Power Centre



Position	Connection	Port Dimension
A	Low Pressure Out (ERD)	3" 3000 PSI CLASS SAE 4-BOLT FLANGE
B	Low Pressure In (Pump)	3" 3000 PSI CLASS SAE 4-BOLT FLANGE
C	High Pressure In (ERD)	3" 3000 PSI CLASS SAE 4-BOLT FLANGE
D	High Pressure Out (Pump)	3" 3000 PSI CLASS SAE 4-BOLT FLANGE
E	Case Drain (Pump)	1 1/2" 3000 PSI CLASS SAE 4-BOLT FLANGE
F	Case Drain (ERD)	1 1/2" 3000 PSI CLASS SAE 4-BOLT FLANGE
G	Inlet Pressure Switch Port	G1/4" BSPP FEMALE

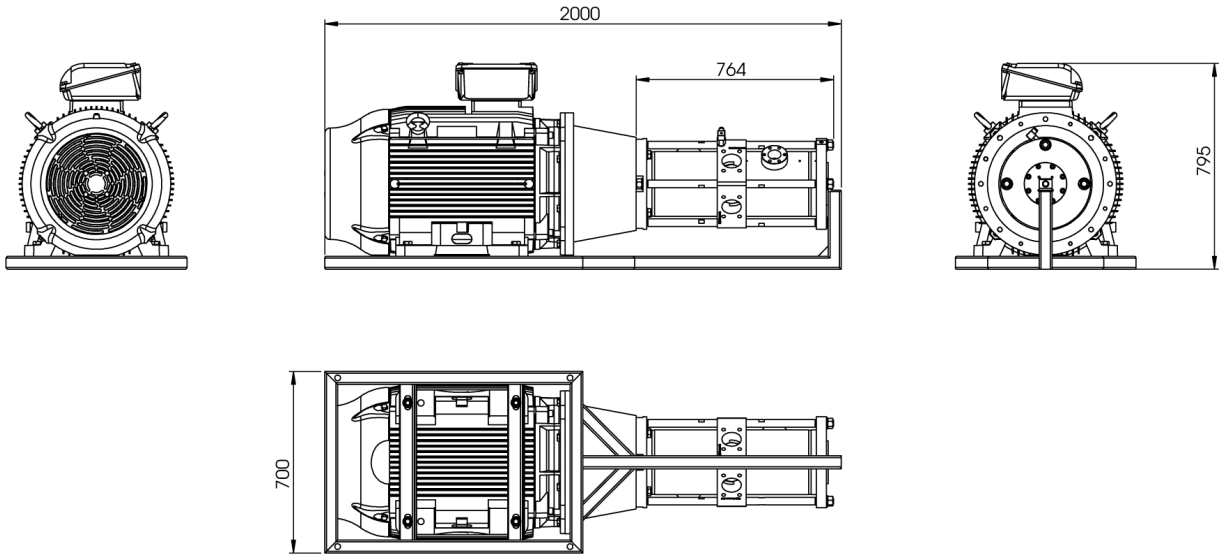
Please note: The rear of the Power centre must be supported on installation.

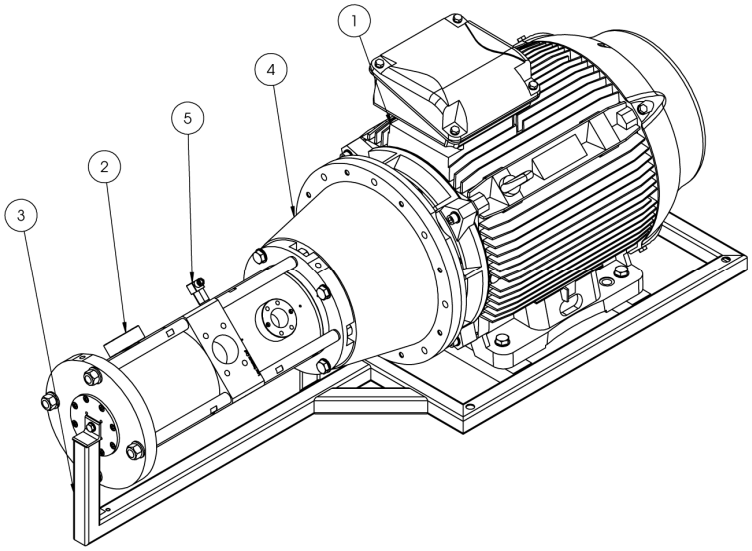
Dimensions and port connection designations are source values. Confirm approved installation envelope and flexible connections for the selected configuration.

PC500 | Assembly & mounting

Manufacturer technical reference | dimensions govern; reproduction is not to scale.

Power Centre Assembly





Item Number	Description	Quantity
1	Electric Motor	1
2	Power Centre	1
3	Mounting Frame	1
4	Bellhousing & Flexible Gear Coupling	1
5	Inlet Protection Pressure Switch Set 3.5 bar	1

Support the unit and motor on the specified assembly. Include alignment, coupling guarding, lifting and maintenance clearances in the approved installation design.

PC500 Power Centre V.01.25.pdf, p. 10.